

Exponential Stability with RISE Controllers for Uncertain Nonlinear Systems with Unknown Time-Varying State Delays

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Abstract—Robust Integral of the Sign of the Error (RISE) controllers have gained popularity in recent years due to their ability to achieve asymptotic tracking error convergence using a continuous control input for uncertain nonlinear systems. In a recent breakthrough, it was shown that the tracking error convergence with RISE controllers is also exponential, whereas previously it was thought to be only asymptotic. However, it remains an open question whether this exponential stability result also holds for systems containing unknown time-varying state delays. In this paper, a novel strict candidate Lyapunov function is developed to prove exponential stability with a RISE controller for uncertain nonlinear systems involving unknown time-varying state delays. Additionally, a simulation example is provided to demonstrate the performance of a RISE controller in the presence of unknown time-varying state delays. The results indicate a higher sensitivity of tracking error and control effort to the delay frequency as compared to the delay magnitude.

I. INTRODUCTION

Robust Integral of the Sign of the Error (RISE) controllers have gained popularity in recent years due to their ability to achieve asymptotic tracking error convergence using a continuous control input for uncertain nonlinear systems with bounded exogenous disturbances [1]–[3]. The success of RISE controllers is powered by their ability to implicitly identify and reject time-varying additive exogenous disturbances or modeling uncertainties. Despite the popularity of RISE controllers, the best stability result that was known for RISE controllers was asymptotic tracking error convergence, until the recent work [4] was published.

In [4], it was shown that, in addition to being asymptotic, the tracking error convergence with RISE controllers is also uniform and exponential. The exponential stability result in [4] is achieved by constructing a novel strict Lyapunov function, which consists of standard sum of norm squared error terms and an auxiliary term called the P-function. Unlike previous results for RISE controllers [1], the P-function in [4] is designed such that it yields a strict Lyapunov function.

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Although the result in [4] guarantees exponential stability with RISE controllers for systems involving exogenous disturbances and modeling uncertainties, an open question is whether the exponential stability result also holds for uncertainties involving unknown time-varying state delays.

Time delays exist in many practical engineering systems [5]. In particular, time delays in the state appear in applications such as remote teleoperation of machines. Numerous results (cf., [6]–[13]) develop controllers to reject uncertainties involving state delays. In such results, the perturbing delay effect is addressed using a constructive Lyapunov-based analysis in which Lyapunov-Krasovskii (LK) functionals are used. LK functionals are typically used to eliminate delay-dependent terms in the subsequent Lyapunov-based stability analysis.

The results in [11] and [12] use a RISE-based controller to compensate for state-delayed uncertainties. However, the candidate Lyapunov function used in these results is non-strict, because when these results were published, it was not clear how an appropriate P-function and an LK functional can be constructed to yield a strict candidate Lyapunov function, and hence, only asymptotic tracking error convergence was proven.

In this paper, a strict candidate Lyapunov function is constructed to prove exponential stability with a RISE controller for uncertain nonlinear systems with additive exogenous disturbances and unknown time-varying state delays. In addition to the standard sum of norm squared error terms, the constructed candidate Lyapunov function consists of an appropriate P-function and an LK functional. The P-function and the LK functional are designed in tandem to eliminate the state-dependent and delayed-state-dependent uncertain terms in the ensuing stability analysis, while also yielding a negative definite time-derivative of the candidate Lyapunov function. Consequently, an exponential stability result is obtained. Simulation results are provided to demonstrate the performance of the developed controller.

II. CONTROL DESIGN

A. Preliminaries

The Lebesgue measure on $\mathbb{R}^n$ is denoted by μ . The notation C^m denotes the space of continuous functions with continuous first m derivatives. The p -norm is denoted by

$\|\cdot\|_p$, and $\|\cdot\|$ denotes the Euclidean norm. Given some sets A and B , a set-valued map F from A to subsets of B is denoted by $F : A \rightrightarrows B$. The notation $\overline{\text{co}}A$ denotes the closed convex hull of the set A . The notation $B(x, \delta)$, for $x \in \mathbb{R}^n$ and $\delta > 0$, is used to denote the set $\{y \in \mathbb{R}^n : \|x - y\| < \delta\}$. Consider a Lebesgue measurable and locally essentially bounded function $h : \mathbb{R}^n \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^n$. The Filippov regularization of h is defined as $K[h](y, t) \triangleq \bigcap_{\delta > 0} \bigcap_{\mu S_m = 0} \overline{\text{co}} h(B(y, \delta) \setminus S_m, t)$, where $\bigcap_{\mu S_m = 0}$ denotes the intersection of all sets S_m of

Lebesgue measure zero [14, Equation 2b]. The notation $\stackrel{a.a.t.}{\cdot}$ implies that the relation $(\cdot)$ holds for almost all $t \in \mathcal{I}$, given some interval $\mathcal{I}$. A function $y : \mathcal{I}_y \rightarrow \mathbb{R}^n$ is called a Filippov solution of $\dot{y} = h(y, t)$ on the interval $\mathcal{I}_y \subseteq \mathbb{R}_{\geq 0}$, if y is absolutely continuous on $\mathcal{I}_y$, and is a solution to the differential inclusion $\dot{y} \stackrel{a.a.t.}{\in} K[h](y, t)$. The gradient operator is denoted by ∇ , and Clarke's generalized gradient for a locally Lipschitz function $V : \mathbb{R}^n \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}$ is defined as $\partial V(x, t) \triangleq \overline{\text{co}}\{\lim \nabla V(x, t) | (x_i, t_i) \rightarrow (x, t), (x_i, t_i) \notin \Omega_V\}$, where Ω_V denotes the set of measure zero wherever ∇V is not defined [15, Def. 2.2].

B. Control Objective

Consider the nonlinear control affine system

$$\dot{x} = f(x, t) + g(x(t - \tau(t)), t) + d(t) + u, \quad (1)$$

where $x : \mathcal{I} \rightarrow \mathbb{R}^n$ denotes a solution to (1), with the interval of existence $\mathcal{I} = [t_0, t_1]$ with some $t_0, t_1 \in \mathbb{R}_{\geq 0}$. Additionally, $f, g : \mathbb{R}^n \times [0, \infty) \rightarrow \mathbb{R}^n$ are unknown continuously differentiable functions that represent modeling uncertainty in the system, $\tau : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ denotes the unknown time-varying state delay, $d : [0, \infty) \rightarrow \mathbb{R}^n$ denotes an exogenous $\mathcal{C}^2$ disturbance, and $u : \mathcal{I} \rightarrow \mathbb{R}^n$ represents the control input. In the subsequent development, a time-delayed function is denoted as

$$\zeta_\tau \triangleq \begin{cases} \zeta(t - \tau) & t - \tau > t_0 \\ 0 & t - \tau \leq t_0. \end{cases}$$

Let the tracking error $e : \mathcal{I} \rightarrow \mathbb{R}^n$ be defined as

$$e \triangleq x - x_d, \quad (2)$$

where $x_d : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}^n$ is a $\mathcal{C}^2$ reference trajectory. The control objective is to prove exponential stability of the tracking error system despite the exogenous disturbance and uncertain time-varying state delay. Similar to [11]–[13], the following standard assumptions are used in this paper.

Assumption 1. The state delay is bounded as $0 \leq \tau \leq \varphi_1$, and the rate of change of delay is bounded as $|\dot{\tau}| \leq \varphi_2 < 1$, where $\varphi_1, \varphi_2 \in \mathbb{R}_{>0}$ are known positive constants.

Assumption 2. The functions f and g and their first and second partial derivatives are bounded on each subset of their domain of the form $\mathcal{K} \times [0, \infty)$, where $\mathcal{K} \subset \mathbb{R}^n$ is compact. Additionally, the bound corresponding to $\mathcal{K}$ is known.

Assumption 3. The first two derivatives of the disturbance term d are bounded by known constants.

Assumption 4. The reference trajectory x_d and its first two derivatives are bounded by known constants.

C. Control Law Development

Let $r : \mathcal{I} \rightarrow \mathbb{R}^n$ denote an unmeasurable auxiliary tracking error defined as $r \triangleq \dot{e} + \alpha e$, with some user-selected constant gain $\alpha \in \mathbb{R}_{>0}$. Consequently, the tracking error dynamics can be rewritten as

$$\dot{e} = r - \alpha e. \quad (3)$$

Let $z : \mathcal{I} \rightarrow \mathbb{R}^{2n}$ be defined as $z \triangleq \begin{bmatrix} e^T & r^T \end{bmatrix}^T$. Based on the subsequent stability analysis, a continuous RISE control input is designed as [1]

$$u \triangleq -\alpha e - \hat{S}. \quad (4)$$

In (4), $\hat{S} : \mathcal{I} \rightarrow \mathbb{R}^n$ is designed as

$$\begin{aligned} \hat{S}(t) &= \hat{S}(t_0) + ke(t) - ke(t_0) \\ &\quad + \int_{t_0}^t ((k\alpha + 1)e(\sigma) + \beta \text{sgn}(e(\sigma))) d\sigma, \end{aligned} \quad (5)$$

where $k, \beta \in \mathbb{R}_{>0}$ are constant gains, and the initial condition $\hat{S}(t_0)$ can be arbitrarily selected by the user. The Riemann integrability of the term $\beta \text{sgn}(e(\cdot))$ is guaranteed on $[t_0, t]$, $\forall t \in \mathcal{I}$ by [4, Lemma 2]. Taking the time-derivative on both sides in (5) yields that $\hat{S}$ is a solution to the differential inclusion

$$\dot{\hat{S}} \stackrel{a.a.t.}{\in} kr + e + \beta K[\text{sgn}(e)]. \quad (6)$$

Using (1)–(4) yields

$$r = f(x, t) + g(x_\tau, t) + d(t) - \dot{x}_d - \hat{S}. \quad (7)$$

Let $S_1 \triangleq f(x, t) - f(x_d, t) + g(x_\tau, t) - g(x_{d\tau}, t)$ and $S_2 \triangleq f(x_d, t) + g(x_{d\tau}, t) + d(t) - \dot{x}_d$. Then, (7) can be rewritten as

$$r = S_1 + S_2 - \hat{S}. \quad (8)$$

Based on (6) and (8), r is a solution to the differential inclusion

$$\dot{r} \stackrel{a.a.t.}{\in} \tilde{N} + N_B - kr - e - \beta K[\text{sgn}](e), \quad (9)$$

where $\tilde{N} \triangleq \dot{S}_1$ and $N_B \triangleq \dot{S}_2$. Based on Assumptions 2 and 3, it can be shown that $\|N_B\| \leq \gamma_1$ and $\|\dot{N}_B\| \leq \gamma_2$, where $\gamma_1, \gamma_2 \in \mathbb{R}$ are known positive constants. The Mean Value Theorem (MVT)-based inequality in [16, Lemma 2] can be utilized to find a bound on $\tilde{N}$ as

$$\|\tilde{N}\| \leq \rho_1 (\|z\|) \|z\| + \rho_2 (\|z_\tau\|) \|z_\tau\|, \quad (10)$$

where the bounding terms $\rho_1, \rho_2 : [0, \infty) \rightarrow [0, \infty)$ are positive, strictly increasing, and radially unbounded functions. The bound on $\tilde{N}$ in (10) contains separated delay-dependent and delay-free bounding functions.

III. STABILITY ANALYSIS

In [4], RISE controllers are shown to yield exponential tracking error convergence for certain classes of uncertain nonlinear systems. Such a stability result is achieved by developing an auxiliary function known as the P-function in the stability analysis. However, the result in [4] does not apply for state-delayed systems. To guarantee exponential tracking error convergence for the state-delayed system in (1), we use a P-function and an LK functional in tandem in the stability analysis. For the closed-loop error system in (3) and (9), and based on the subsequent stability analysis, we design the P-function $P : \mathcal{I} \rightarrow \mathbb{R}_{>0}$ as

$$P \triangleq \beta \|e\|_1 - e^T N_B + e^{-\lambda_P t} * (e^T \dot{N}_B) + e^{-\lambda_P t} * ((\alpha - \lambda_P) (\beta \|e\|_1 - e^T N_B)), \quad (11)$$

where $\lambda_P \in \mathbb{R}_{>0}$ is a user-selected constant, and the notation $*$ denotes the convolutional integral $e^{-\lambda_P t} * q = \int_{t_0}^t e^{-\lambda_P(t-\sigma)} q(\sigma) d\sigma$, for any given $q : [t_0, \infty) \rightarrow \mathbb{R}$. It can be verified using the Leibniz rule that $\frac{d}{dt} \left(\int_{t_0}^t e^{-\lambda_P(t-\sigma)} q(\sigma) d\sigma \right) = q(t) - \lambda_P \int_{t_0}^t e^{-\lambda_P(t-\sigma)} q(\sigma) d\sigma$. Therefore, the convolutional integral satisfies the property $\frac{d}{dt} (e^{-\lambda_P t} * q) = q(t) - \lambda_P e^{-\lambda_P t} * q$. Since $t \mapsto e(t)$ is absolutely continuous and $\|\cdot\|_1$ is globally Lipschitz, the mapping $t \mapsto \|e(t)\|_1$ is differentiable for almost all time. Therefore, using the chain rule in [15, Theorem 2.2] yields $\frac{d}{dt} (\|e\|_1) \stackrel{a.a.t.}{\in} K[\text{sgn}](e)$. Therefore, taking the time-derivative of (11), using Leibniz's rule, and substituting (3) and (11) into the resulting time-derivative yields¹

$$\dot{P} \stackrel{a.a.t.}{\in} -\lambda_P P + r^T \beta K[\text{sgn}](e) - r^T N_B. \quad (12)$$

Moreover, provided the gain conditions

$$\alpha > \lambda_P \quad (13)$$

and

$$\beta > \gamma_1 + \frac{\gamma_2}{\alpha - \lambda_P} \quad (14)$$

are satisfied, it can be shown using [4, Lemma 4] that $P \geq 0$, $\forall t \in \mathcal{I}$. To eliminate the delay-dependent terms in the subsequent stability analysis, we introduce the LK functional $Q : \mathcal{I} \rightarrow \mathbb{R}_{>0}$ defined as

$$Q(t) = \int_{t-\tau(t)}^t e^{-\lambda_Q(t-\sigma)} \rho_2^2(\|z(\sigma)\|) \|z(\sigma)\|^2 d\sigma, \quad (15)$$

where $\lambda_Q \in \mathbb{R}_{>0}$ is a user-selected constant. Using Leibniz rule, the time-derivative of the LK functional is obtained as

$$\dot{Q} = -\lambda_Q Q + \rho_2^2(\|z(t)\|) \|z(t)\|^2 - e^{-\lambda_Q \tau} (1 - \dot{\tau}) \rho_2^2(\|z_\tau\|) \|z_\tau\|^2. \quad (16)$$

¹The reader is referred to [4, Proof of Lemma 3] for a more detailed derivation.

Let $\psi \triangleq [e^T \ r^T \ P \ Q]^T$. Furthermore, let $V_L : \mathbb{R}^{2n+2} \rightarrow \mathbb{R}_{\geq 0}$ be defined as

$$V_L(\psi) \triangleq \frac{1}{2} e^T e + \frac{1}{2} r^T r + P + Q. \quad (17)$$

Additionally, let $\gamma_3, k_{\min} \in \mathbb{R}_{>0}$ and $\rho, \bar{\rho} : [0, \infty) \rightarrow [0, \infty)$ be defined as

$$\gamma_3 \triangleq \frac{e^{\lambda_Q \varphi_1}}{4(1 - \varphi_2)}, \quad (18)$$

$$k_{\min} \triangleq \min\{k - \gamma_3, \alpha, \frac{\lambda_P}{2}, \frac{\lambda_Q}{2}\}, \quad (19)$$

$$\rho(\cdot) \triangleq \rho_1(\cdot) + \rho_2^2(\cdot), \quad (20)$$

and

$$\bar{\rho}(\cdot) \triangleq \rho(\cdot) - \rho(0). \quad (21)$$

Furthermore, let $\lambda_V \in \mathbb{R}_{>0}$ denote an arbitrary user-selected constant.

Theorem 1. Let $W : \mathbb{R}^{2n} \rightarrow \mathbb{R}_{\geq 0}$ be defined as $W(z(t_0)) \triangleq \sqrt{\|z(t_0)\|^2 + 2(\beta + \gamma_1) \|z(t_0)\|_1}$. Given any initial condition $z(t_0) \in \mathbb{R}^{2n}$, all solutions to (3) and (9) satisfy $\|z(t)\| \leq W(z(t_0)) e^{-\lambda_V(t-t_0)}$, $\forall (z(t_0), t) \in \mathbb{R}^{2n} \times [t_0, \infty)$, provided that the gains α, β , and λ_P are selected according to the gain conditions in (13) and (14), and the gain k is selected to satisfy

$$k_{\min} > \lambda_V + \rho(W(z(t_0))). \quad (22)$$

Proof: Let $G : \mathbb{R}^{2n+2} \rightrightarrows \mathbb{R}^{2n+2}$ denote the set-valued map

$$G(\psi, t) \triangleq \begin{bmatrix} r - \alpha e \\ -kr - e - \beta K[\text{sgn}](e) + N_B + \tilde{N} \\ -\lambda_P P + r^T \beta K[\text{sgn}](e) - r^T N_B \\ \left(\begin{array}{c} -\lambda_Q Q + \rho_2^2(\|z(t)\|) \|z(t)\|^2 \\ -e^{-\lambda_Q \tau} (1 - \dot{\tau}) \rho_2^2(\|z_\tau\|) \|z_\tau\|^2 \end{array} \right) \end{bmatrix}.$$

Using (3), (9), (12), and (16) yields $\dot{\psi} \stackrel{a.a.t.}{\in} G(\psi, t)$. Based on [15, Theorem 2.2], the time-derivative of V_L along the trajectories $t \mapsto \psi(t)$ exists for almost all time, and

$$\begin{aligned} \dot{V}_L(\psi, t) &\stackrel{a.a.t.}{\in} \bigcap_{\xi \in \partial V_L(\psi)} \xi^T G(\psi, t) \\ &= \nabla V_L^T G(\psi, t) \\ &= [z^T \ 1 \ 1]^T G(\psi, t). \end{aligned} \quad (23)$$

In (23), $\partial V_L(\psi)$ denotes Clarke's generalized gradient. Since the gradient of $V_L(\psi)$, i.e., $\nabla V_L(\psi)$, exists and is continuous for all $\psi \in \mathbb{R}^{2n+2}$, $\partial V_L = \{\nabla V_L\}$. Evaluating (23) yields

$$\begin{aligned} \dot{V}_L &\stackrel{a.a.t.}{\in} r^T (-kr - e - \beta K[\text{sgn}](e) + N_B + \tilde{N}) \\ &\quad - r^T N_B + e^T (r - \alpha e) + r^T \beta K[\text{sgn}](e) \\ &\quad - \lambda_Q Q - \lambda_P P + \rho_2^2(\|z\|) \|z\|^2 \\ &\quad - e^{-\lambda_Q \tau} (1 - \dot{\tau}) \rho_2^2(\|z_\tau\|) \|z_\tau\|^2. \end{aligned} \quad (24)$$

Based on [4, Lemma 1], the set of time-instants where the term $r^T K [\text{sgn}](e(\cdot))$ is set-valued has Lebesgue measure zero. Therefore,

$$\begin{aligned} \dot{V}_L &\stackrel{a.a.t.}{\leq} -\lambda_P P - \alpha \|e\|^2 - \lambda_Q Q - k \|r\|^2 \\ &\quad + \rho_1 (\|z\|) \|z\| \|r\| + \rho_2^2 (\|z\|) \|z\|^2 \\ &\quad + \rho_2 (\|z_\tau\|) \|z_\tau\| \|r\| \\ &\quad - e^{-\lambda_Q \tau} (1 - \dot{\tau}) \rho_2^2 (\|z_\tau\|) \|z_\tau\|^2. \end{aligned} \quad (25)$$

Using Young's inequality yields

$$\begin{aligned} \rho_2 (\|z_\tau\|) \|z_\tau\| \|r\| &\leq e^{-\lambda_Q \tau} (1 - \dot{\tau}) \rho_2^2 (\|z_\tau\|) \|z_\tau\|^2 \\ &\quad + \frac{e^{\lambda_Q \tau} \|r\|^2}{4(1 - \dot{\tau})}. \end{aligned} \quad (26)$$

Using Assumption 1, the term $\frac{e^{\lambda_Q \tau}}{4(1 - \dot{\tau})}$ can be bounded by a constant $\frac{e^{\lambda_Q \tau}}{4(1 - \dot{\tau})} \leq \frac{e^{\lambda_Q \varphi_1}}{4(1 - \varphi_2)} = \gamma_3$ defined in (18). Substituting (26) into (24), and using $\|r\| \leq \|z\|$ yields

$$\begin{aligned} \dot{V}_L &\stackrel{a.a.t.}{\leq} -\lambda_P P - \lambda_Q Q - (k - \gamma_3) \|r\|^2 - \alpha \|e\|^2 \\ &\quad + (\rho_1 (\|z\|) + \rho_2^2 (\|z\|)) \|z\|^2. \end{aligned} \quad (27)$$

Substituting (18)-(21) into (27), and using the inequality $\|z\| \leq \sqrt{\|z^2\| + 2P + 2Q} = \sqrt{2V_L}$ yields

$$\begin{aligned} \dot{V}_L &\stackrel{a.a.t.}{\leq} -2(k_{\min} - \rho(\|z\|)) V_L \\ &\leq -2(k_{\min} - \rho(\sqrt{2V_L})) V_L \\ &= -2(k_{\min} - (\bar{\rho}(\sqrt{2V_L}) + \rho(0))) V_L. \end{aligned} \quad (28)$$

Consider the open and connected set $\mathcal{D} \triangleq \{\sigma \in \mathbb{R}^{2n+2} | V_L(\sigma) < \frac{(\bar{\rho}^{-1}(k_{\min} - \rho(0) - \lambda_V))^2}{2}\}$. Based on (28), V_L is non-increasing along all trajectories initialized such that $\psi(t_0) \in \mathcal{D}$, which is ensured by selecting

$$\begin{aligned} k_{\min} &> \lambda_V + \bar{\rho}(\sqrt{2V_L(\psi(t_0))}) + \rho(0) \\ &= \lambda_V + \rho(\sqrt{2V_L(\psi(t_0))}). \end{aligned} \quad (29)$$

Substituting $P(t_0) = \beta \|e(t_0)\|_1 - e^T(t_0)N_B(t_0)$ and $Q(t_0) = 0$ yields $V_L(\psi(t_0)) = \frac{1}{2} \|z(t_0)\|^2 + (\beta \|e(t_0)\|_1 - e^T(t_0)N_B(t_0))$. Although the term $N_B(t_0)$ is unknown, it can be bounded as $\|N_B(t_0)\| \leq \gamma_1$; hence, using the property of norms, $\|e(t_0)\| \leq \|e(t_0)\|_1 \leq \|z(t_0)\|_1$, it follows that

$$\begin{aligned} \sqrt{2V_L(\psi(t_0))} &\leq \sqrt{\|z(t_0)\|^2 + 2(\beta + \gamma_1) \|z(t_0)\|_1} \\ &= W(z(t_0)), \end{aligned} \quad (30)$$

where $W(\cdot)$ is defined in the theorem statement. Consequently, selecting $k_{\min}$ according to (22) guarantees that (29) is satisfied. Therefore, it follows from (28) that selecting the gains k , α , λ_P , and λ_Q such that $k_{\min}$ satisfies the gain condition in (22) yields

$$\dot{V}_L \stackrel{a.a.t.}{\leq} -2\lambda_V V_L. \quad (31)$$

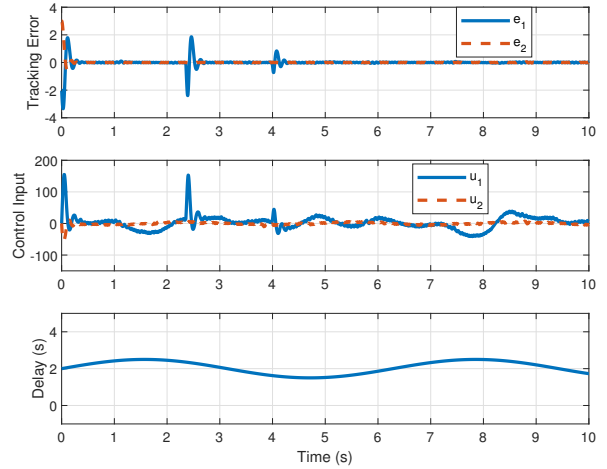


Figure 1. Plots of tracking error, control input, and state delay with respect to time for Case II.

Based on the Comparison Principle [17, Lemma 4.4], it follows that

$$V_L(\psi(t)) \leq V_L(\psi(t_0))e^{-2\lambda_V(t-t_0)}, \forall t \in \mathcal{I}. \quad (32)$$

Using (32), $\|z(t)\|$ can further be upper-bounded as

$$\|z(t)\| \leq \sqrt{2V_L(\psi(t_0))}e^{-\lambda_V(t-t_0)}, \forall t \in \mathcal{I}. \quad (33)$$

Additionally, since $(\psi, t) \mapsto G(\psi, t)$ is a locally bounded mapping, and ψ is precompact (i.e., bounded over any interval $\mathcal{I}$) according to (33), invoking [18, Lemma 3.3 and Remark 3.4] rules out the possibility of solutions escaping in finite time. Therefore, $\mathcal{I} = [t_0, \infty)$. Thus, the exponential convergence in (33) holds for all $t \in [t_0, \infty)$. Therefore, substituting (30) into (33) yields

$$\|z(t)\| \leq W(z(t_0))e^{-\lambda_V(t-t_0)}, \forall t \in [t_0, \infty). \quad (34)$$

Remark 1. It follows from (8) that r is the estimation error between the state-delayed uncertainty $S_1 + S_2$ and the RISE term $\hat{S}$. Therefore, the implicit learning property of RISE controllers with an exponential rate of convergence is retained even when the uncertainty consists of delayed state-dependent terms with unknown delays.

Remark 2. In [4, Eq. 11], the initial condition for P is incorrectly initialized as $P(t_0) = 0$. This error is easily corrected by using $P(t_0) = \beta \|e(t_0)\|_1 - e^T(t_0)N_B(t_0)$. Consequently, [4, Equations 16 and 23] can be replaced by equations (22) and (33) as in this paper.

IV. SIMULATION RESULTS

To demonstrate the performance of a RISE controller, we provide simulation results for a state-delayed Van der Pol

Table I
RMS ERROR AND CONTROL EFFORT FOR TIME-VARYING STATE
DELAYS

	τ (seconds)	$\ e_{\text{RMS}}\ $	$\ u_{\text{RMS}}\ $
Case I	$1 + 0.5 \sin(t)$	0.356	23.251
Case II	$2 + 0.5 \sin(t)$	0.370	23.127
Case III	$2 + 0.5 \sin(2t)$	0.335	20.090
Case IV	$2 + \sin(2t)$	0.321	22.329
Case V	$2 + \sin(5t)$	0.352	26.000
Case VI	$2 + \sin(20t)$	0.542	30.274
Case VII	$2 + 2 \sin(20t)$	0.528	30.389

oscillator given by the dynamics in (1) with

$$\begin{aligned} f(x) &\triangleq \begin{bmatrix} 10 \left(x_1 - \frac{1}{3}x_1^3 - x_2 \right) \\ \frac{1}{10}x_1 \end{bmatrix}, \\ g(x_\tau) &\triangleq \begin{bmatrix} 10 \left(x_{\tau 1} - \frac{1}{3}x_{\tau 1}^3 - x_{\tau 2} \right) \\ \frac{1}{10}x_{\tau 1} \end{bmatrix}, \end{aligned}$$

where $x = [x_1, x_2]^T$, $x_\tau = [x_{\tau 1}, x_{\tau 2}]^T$. Additionally, the disturbance $d(t)$ is modeled as an additive white Gaussian (AWG) signal with a zero mean and a standard deviation of 1. The desired trajectory is $x_d = [\sin(5t), 2 \cos(2t)]$. The system is initialized as $x(0) = [-2, 5]^T$ and $\hat{S}(0) = 0$. The control gains are selected as $k = 30$, $\alpha = 30$, and $\beta = 200$.

Table I provides the root mean squared (RMS) values of tracking error and control effort, denoted by e_{RMS} and u_{RMS} , respectively, for various cases of time-varying state-delays with different magnitudes and frequencies. The results indicate a higher sensitivity of tracking error and control effort to the delay frequency as compared to the delay magnitude. Figure 1 shows the plots of the tracking error, control input, and state delay for Case II. The tracking error is seen converging to zero.

V. CONCLUSION

In this paper, a novel strict candidate Lyapunov function is developed to prove exponential stability with a RISE controller for uncertain nonlinear systems with unknown time-varying state delays. Additionally, a simulation example is provided to demonstrate the performance of a RISE controller in presence of unknown time-varying state delays. The results indicate a higher sensitivity of tracking error and control effort to the delay frequency as compared to the delay magnitude. Future work may involve extension of the results in this paper for the more challenging problem of designing a RISE controller and proving exponential stability for systems with time-delays in the control input. Additionally, the developments in this paper can be leveraged for application to remote teleoperation applications with exponential stability guarantees and improved transparency to the teleoperator.

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